Thermodynamic Limit and Basin Scaling in the LSR Energy

Analysis of Hoover et al. (ICLR 2025 Workshop)

1 Setup and Notation

The LSR (log-sum-ReLU) energy function introduced in [1] is

$$E_{\mathrm{LSR}}(\mathbf{x}) = -\frac{1}{\beta}\log\left(\epsilon + \sum_{\mu=1}^{M}\mathrm{ReLU}\left(1 - \frac{\beta}{2}\|\mathbf{x}-\boldsymbol{\xi}_\mu\|^2\right)\right), \tag{1}$$

where $\boldsymbol{\xi}_\mu \in \{-1,+1\}^D$ are stored memories, $\beta > 0$ is the inverse temperature, and $\epsilon \geq 0$ is a regularization constant.

The support region around memory μ is $\mathrm{S}_\mu = \{\mathbf{x} : \|\mathbf{x}-\boldsymbol{\xi}_\mu\| \leq \sqrt{2/\beta}\}$. Outside $\bigcup_\mu \mathrm{S}_\mu$ (with $\epsilon = 0$) the energy is infinite and the gradient vanishes.

For comparison, the standard LSE (log-sum-exp) energy is

$$E_{\mathrm{LSE}}(\mathbf{x}) = -\frac{1}{\beta}\log\sum_{\mu=1}^{M}\exp\left(-\frac{\beta}{2}\|\mathbf{x}-\boldsymbol{\xi}_\mu\|^2\right). \tag{2}$$

2 Recap of the Main Theorems and Their Intuition

We first summarize the three theorems of [1] and explain what they say in plain terms.

2.1 Theorem 1: Exact retrieval from basins

Statement (paraphrased). Let $r = \min_{\mu\neq\nu}\|\boldsymbol{\xi}_\mu - \boldsymbol{\xi}_\nu\|$ be the minimum distance between any two memories. For basin radius $\Delta \in (0, r)$ and $\beta = 2/(r-\Delta)^2$, every input $\mathbf{x} \in S_\mu(\Delta) \equiv \{\mathbf{x} : \|\mathbf{x}-\boldsymbol{\xi}_\mu\| \leq \Delta\}$ is retrieved exactly as $\boldsymbol{\xi}_\mu$ via energy gradient descent. With an appropriate learning rate, a *single* gradient step suffices.

Intuition. The ReLU separation function has finite reach: memory μ only influences points within distance $\sqrt{2/\beta}$ from it. If β is large enough, this reach is $\sqrt{2/\beta} = r - \Delta < r$. So any point at distance $\leq \Delta$ from memory μ is within the support of μ but *outside* the support of every other memory (since the nearest other memory is at distance $\geq r$ away, and $\Delta + (r-\Delta) = r$). The gradient then points purely toward $\boldsymbol{\xi}_\mu$, with no interference from other memories.

Hidden constraint: $\Delta \leq r/2$. Note a subtlety glossed over in [1]. The proof claims that for $\mathbf{x} \in S_\mu(\Delta)$, we have $B(\mathbf{x}) = \{\mu\}$, meaning $\mathbf{x}$ is in the support of memory μ. This requires $\|\mathbf{x}-\boldsymbol{\xi}_\mu\| \leq \sqrt{2/\beta} = r - \Delta$. But since $\|\mathbf{x}-\boldsymbol{\xi}_\mu\| \leq \Delta$ by definition of the basin, we need $\Delta \leq r - \Delta$, i.e.,

$$\Delta \;\leq\; \frac{r}{2}. \tag{3}$$

If $\Delta > r/2$, then points at the outer edge of the "basin" are *outside* the support of their own memory — they sit in a dead zone with zero gradient. So the effective basin can never exceed half the inter-memory distance.

2.2 What happens when $\beta \neq 2/(r-\Delta)^2$?

The prescribed value $\beta = 2/(r-\Delta)^2$ is a knife-edge. It is instructive to consider what happens on either side, since the paper does not discuss this. The three regimes are controlled by the support radius $R = \sqrt{2/\beta}$:

Case 1: $\beta > 2/(r-\Delta)^2$ ("too cold", $R < r - \Delta$). The support radius shrinks below $r - \Delta$. This is *more* than enough to isolate each memory (neighboring supports are still disjoint), so there is no interference. However, the effective basin of attraction also shrinks: it is no longer Δ but the support radius $R = \sqrt{2/\beta}$ itself. Any point farther than R from all memories is in a dead zone.

In this regime:

- Retrieval is exact for $\|\mathbf{x} - \boldsymbol{\xi}_\mu\| \leq R$,
- but the basin is *smaller* than desired ($R < r - \Delta < \Delta$ for Δ near $r/2$),
- and a larger fraction of space is dead (zero gradient).

Case 2: $\beta = 2/(r-\Delta)^2$ (the Theorem 1 prescription, $R = r - \Delta$). This is the optimal trade-off: the support radius is as large as possible while keeping each basin free of interference from other memories. Every $\mathbf{x} \in S_\mu(\Delta)$ sees exactly one memory, retrieval is exact.

Case 3: $\beta < 2/(r-\Delta)^2$ ("too hot", $R > r - \Delta$). The support balls now overlap: a point $\mathbf{x}$ near memory μ may also fall within the support of a neighboring memory ν (specifically, whenever $\|\mathbf{x} - \boldsymbol{\xi}_\nu\| \leq R$ while $\|\boldsymbol{\xi}_\mu - \boldsymbol{\xi}_\nu\| = r < 2R$). In the overlap region, $|B(\mathbf{x})| > 1$, and the gradient becomes an average over multiple memories. Gradient descent no longer converges to any individual $\boldsymbol{\xi}_\mu$ but instead to the centroid $|B|^{-1} \sum_{\mu \in B} \boldsymbol{\xi}_\mu$ — an *emergent minimum.*

In this regime:

- Original memories are no longer guaranteed to be retrieved from their full basin $S_\mu(\Delta)$,
- but they remain local minima (at $\mathbf{x} = \boldsymbol{\xi}_\mu$ the gradient is still zero),
- and new local minima (centroids) appear in the overlap regions.

This is exactly the regime of Theorem 3.

Summary: the three regimes.

	Too cold $\beta > \frac{2}{(r-\Delta)^2}$	**Knife-edge** $\beta = \frac{2}{(r-\Delta)^2}$	**Too hot** $\beta < \frac{2}{(r-\Delta)^2}$
Support radius R	$< r - \Delta$	$= r - \Delta$	$> r - \Delta$
Supports overlap?	No	No (tangent)	Yes
Effective basin	$R < \Delta$	Δ	$< \Delta$ (contaminated)
Emergent minima	None	None	Yes (centroids)
Dead zone fraction	Larger	Moderate	Smaller

Remark 1 (Monotonicity illusion)**.** *One might expect that increasing β ("lowering temperature") always sharpens retrieval. In LSE this is true: larger β makes the softmax peakier, which improves basin size. In LSR the opposite happens at large β: the support shrinks and the basin* decreases*, because the ReLU simply cuts off rather than smoothly suppressing distant memories. The finite support fundamentally changes the role of temperature.*

2.3 Theorem 2: Exponential memory capacity

Statement (paraphrased). Consider M memories sampled uniformly from $\{-1,+1\}^D$. With probability at least δ, all memories are retrievable as per Theorem 1, provided

$$M = O\Big(\sqrt{\log(1/\delta)}\ \exp(\alpha^2 D/2)\Big), \tag{4}$$

where $\alpha \in (0,1)$ is a parameter controlling the trade-off between capacity and basin size.

What is α? The parameter α is the *maximum cosine similarity* between any pair of stored memories. In the proof (Lemma 1 of [1]), random binary vectors $\boldsymbol{\xi}_\mu, \boldsymbol{\xi}_\nu \in \{-1,+1\}^D$ satisfy

$$\frac{|\langle \boldsymbol{\xi}_\mu, \boldsymbol{\xi}_\nu \rangle|}{\|\boldsymbol{\xi}_\mu\|\,\|\boldsymbol{\xi}_\nu\|} \ \le\ \alpha \tag{5}$$

with high probability (by concentration of measure). This translates into a lower bound on the minimum pairwise Euclidean distance:

$$r = \min_{\mu\neq\nu} \|\boldsymbol{\xi}_\mu - \boldsymbol{\xi}_\nu\| \ \ge\ \sqrt{2D(1-\alpha)}. \tag{6}$$

Smaller α means the memories are more spread out (closer to orthogonal), giving larger r and larger basins, but fewer memories can be packed. Larger α allows more memories at the cost of tighter spacing. The capacity $M \sim \exp(\alpha^2 D/2)$ grows with α: the more overlap you tolerate, the more patterns you can store.

Intuition. Random binary vectors in high dimensions are nearly orthogonal: their cosine similarity concentrates around zero with fluctuations of order $1/\sqrt{D}$. This means $\sim \exp(cD)$ of them can be packed into $\{-1,+1\}^D$ with all pairwise cosines below α. Since they are well-separated, Theorem 1 applies to each.

2.4 Theorem 3: Emergent local minima

Statement (paraphrased). Under the same configuration, for any $\mathbf{x}$ that lies in the support of more than one memory ($|B(\mathbf{x})| > 1$ where $B(\mathbf{x}) = \{\mu : \|\mathbf{x} - \boldsymbol{\xi}_\mu\| \le \sqrt{2/\beta}\}$), the energy gradient descent converges to the centroid $\frac{1}{|B(\mathbf{x})|}\sum_{\mu\in B(\mathbf{x})} \boldsymbol{\xi}_\mu$, which is *not* any stored memory.

Intuition. In the overlap region where two or more support balls intersect, all the corresponding memories pull on the state simultaneously. Since the ReLU separation function is linear (not exponential), no single memory dominates — the gradient is a balanced average. The fixed point is therefore the centroid of the contributing memories: a *new pattern* that was never stored. The total number of such emergent minima (one per non-empty subset of memories with intersecting supports) can be as large as 2^M.

This is impossible in LSE: the exponential separation function always makes one memory dominate the softmax, suppressing the others.

3 Implicit Scaling: β Cannot Be Fixed as $D \to \infty$

3.1 What the theorems require

From Theorem 2, the minimum pairwise distance satisfies $r \ge \sqrt{2D(1-\alpha)}$, and exact retrieval with basin radius $\Delta \in (0, r)$ requires (Theorem 1)

$$\beta \ =\ \frac{2}{(r-\Delta)^2}. \tag{7}$$

3.2 Why Δ must be proportional to r

The basin radius Δ is the maximum corruption from which a memory can be recovered. To be a useful associative memory, Δ must scale with the natural noise level. Consider a stored binary pattern $\boldsymbol{\xi}_\mu \in \{-1,+1\}^D$ corrupted by flipping a fraction f of its bits. The corrupted vector $\tilde{\boldsymbol{\xi}}$ satisfies

$$\|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}_\mu\|^2 = 4fD \quad \Longrightarrow \quad \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}_\mu\| = 2\sqrt{fD}. \tag{8}$$

Since $r \sim \sqrt{2D}$, this corruption distance is $\sim (2\sqrt{f}/\sqrt{2})\, r = \sqrt{2f}\; r$. Retrieving a pattern with fraction f of bits flipped requires $\Delta \geq 2\sqrt{fD} \propto r$. If Δ did not grow with r (and hence with $\sqrt{D}$), the memory could only correct a *vanishing* fraction of bit flips as $D \to \infty$ — an increasingly useless memory.

More directly: the basin must be large enough that no other memory falls inside it, which requires $\Delta < r$, but small enough that the support ball $\sqrt{2/\beta} = r - \Delta$ stays within the exclusive zone. The only consistent choice is $\Delta = \eta\, r$ for some fixed $\eta \in (0,1)$.

3.3 $\beta = O(1/D)$ follows immediately

With $\Delta = \eta\, r$ and $r \sim \sqrt{2D(1-\alpha)}$:

$$\beta \;=\; \frac{2}{(r-\eta r)^2} \;=\; \frac{2}{(1-\eta)^2 r^2} \;\sim\; \frac{2}{(1-\eta)^2 \cdot 2D(1-\alpha)} \;=\; \frac{1}{(1-\eta)^2(1-\alpha)\, D}. \tag{9}$$

More generally, for any fixed ratio $\Delta/r < 1$, we obtain $\beta = O(1/D)$. The inverse temperature must vanish as $D \to \infty$.

This is never stated explicitly in [1], but it is embedded in every theorem.

3.4 Consequence for the support radius

The support radius of each memory is

$$R_{\text{supp}} = \sqrt{2/\beta} = O(\sqrt{D}), \tag{10}$$

which matches the natural length scale of the hypercube $\{-1,+1\}^D$ (where all points lie at distance $\sqrt{D}$ from the origin and typical inter-point distances are $\sim\sqrt{2D}$).

If instead β were held fixed as $D \to \infty$:

- $R_{\text{supp}} = \sqrt{2/\beta} = O(1)$, a fixed constant;
- inter-memory distance $r \sim \sqrt{2D} \to \infty$;
- each memory becomes an isolated island of finite energy in a space where all scales grow as $\sqrt{D}$;
- the fraction of space with any gradient signal goes to zero exponentially.

3.5 Contrast with LSE

For the LSE energy (2), β can remain $O(1)$ as $D \to \infty$. The exponential separation function has infinite support: every point $\mathbf{x} \in \mathbb{R}^D$ receives a nonzero gradient pointing toward the softmax-weighted centroid of the memories. The basins of attraction have radius $O(\sqrt{D})$ even with $\beta = O(1)$. No dead zones exist.

4 The Thermodynamic Limit

4.1 Mean-field scaling

Define a rescaled inverse temperature

$$\tilde{\beta} = \beta D, \qquad \text{so that } \beta = \tilde{\beta}/D, \tag{11}$$

and hold $\tilde{\beta}$ fixed as $D \to \infty$. The energy becomes

$$E_{\mathrm{LSR}}(\mathbf{x}) = -\frac{D}{\tilde{\beta}} \log\left(\epsilon + \sum_{\mu=1}^{M} \mathrm{ReLU}\left(1 - \frac{\tilde{\beta}}{2D}\|\mathbf{x} - \boldsymbol{\xi}_\mu\|^2\right)\right). \tag{12}$$

Since $\|\mathbf{x} - \boldsymbol{\xi}_\mu\|^2 = O(D)$ for typical configurations, the ReLU argument is $1 - O(\tilde{\beta}) = O(1)$, and

$$E_{\mathrm{LSR}} = -\frac{D}{\tilde{\beta}} \log\big(O(1)\big) = O(D). \tag{13}$$

Observation 1 (Extensivity)**.** *The energy is extensive (proportional to system size D), as required for a well-defined thermodynamic limit. The correct limit is*

$$D \to \infty, \qquad \tilde{\beta} = \beta D \text{ fixed}, \qquad M \sim e^{cD} \text{ for some } c > 0.$$

This is analogous to the mean-field scaling $J \sim 1/\sqrt{N}$ in classical spin glass models.

4.2 Comparison of scaling regimes

Quantity	**LSE**	**LSR**
Inverse temperature β	$O(1)$	$O(1/D)$
Rescaled temperature $\tilde{\beta} = \beta D$	grows as D	$O(1)$
Support radius $\sqrt{2/\beta}$	∞ (global)	$O(\sqrt{D})$
Basin radius Δ	$O(\sqrt{D})$	$O(\sqrt{D})$
Memory capacity M	$\sim e^{cD}$	$\sim e^{cD}$
Energy E	$O(D)$	$O(D)$
Dead zones	None	Yes

Table 1: Scaling comparison between LSE and LSR in the $D \to \infty$ limit.

5 The Basin Shrinkage Problem

5.1 Absolute vs. relative basin size

In absolute terms, the basin radius $\Delta \sim \sqrt{D}$ *grows* with dimension. The support radius $\sqrt{2/\beta} \sim \sqrt{D}$ also grows. All natural length scales (r, Δ, R_{supp}, $\|\boldsymbol{\xi}_\mu\|$) are $O(\sqrt{D})$, so the geometry is self-similar across dimensions.

In relative terms, however, the *volume fraction* of a D-dimensional ball of radius $c\sqrt{D}$ inside the hypercube $[-1,1]^D$ behaves as

$$\frac{\mathrm{Vol}(B_{c\sqrt{D}})}{\mathrm{Vol}([-1,1]^D)} \sim \left(\frac{\pi e\, c^2}{4}\right)^{D/2} \cdot \text{(subexponential prefactor)}. \tag{14}$$

For moderate c this ratio decays exponentially with D. But this is equally true for LSE — the basins of attraction for LSE are also balls of radius $O(\sqrt{D})$ covering an exponentially small fraction of the space.

5.2 The qualitative difference: dead zones

The fundamental distinction is not basin size but what happens *outside* all basins:

	LSE	**LSR**
Outside all basins	Gradient exists	**Gradient is zero**
Random initialization	Always converges	May land in dead zone
Retrieval from corruption	Works if $\|\delta\| < \Delta$	Same condition

For the **retrieval task** (starting from a corrupted version of a stored memory with bounded noise), both models work identically well: the corrupted input is at distance $O(\sqrt{D})$ from the true memory, which lies within the basin.

For **random initialization** or unconstrained generation, LSR has a fundamental problem: any initial point outside $\bigcup_\mu \mathrm{S}_\mu$ has zero gradient and will not move under gradient descent. The model is blind in these regions.

5.3 Emergent minima in the thermodynamic limit

With $\beta = \tilde{\beta}/D$ and $\tilde{\beta}$ fixed, the support radius $R_{\text{supp}} = \sqrt{2D/\tilde{\beta}}$ and the inter-memory distance $r \sim \sqrt{2D}$. Two memories at distance r have overlapping support regions whenever $2R_{\text{supp}} > r$, i.e.,

$$2\sqrt{2D/\tilde{\beta}} > \sqrt{2D} \quad \Longleftrightarrow \quad \tilde{\beta} < 4. \tag{15}$$

So for $\tilde{\beta} < 4$, supports overlap, emergent minima (centroids of memory subsets) appear, and their number can be as large as 2^M. The basin radius for exact retrieval of the original memories requires $\tilde{\beta} > 2/(1-\Delta/r)^2 \cdot (1/D) \cdot D = 2D/(r-\Delta)^2$; substituting $r \sim \sqrt{2D}$ and $\Delta = \eta r$ for $\eta < 1$:

$$\tilde{\beta} \;\geq\; \frac{1}{(1-\eta)^2(1-\alpha)}. \tag{16}$$

Thus the regime of simultaneous memorization and generation is

$$\frac{1}{(1-\eta)^2(1-\alpha)} \;\leq\; \tilde{\beta} \;<\; 4, \tag{17}$$

which exists for sufficiently small η and α.

6 Summary

1. β **is not fixed.** The theorems in [1] implicitly require $\beta = O(1/D)$. At fixed β, the support regions become negligibly small relative to the ambient space as $D \to \infty$.

2. **A thermodynamic limit exists** with mean-field scaling $\tilde{\beta} = \beta D$ held fixed. The energy is extensive, $E = O(D)$, and memory capacity remains exponential.

3. **Basins grow as $\sqrt{D}$** in absolute terms, matching all other length scales. The exponential shrinkage of volume fractions affects both LSE and LSR equally.

4. **Dead zones are the real cost** of finite support. Unlike LSE, which provides a gradient signal everywhere, LSR has regions of zero gradient that cover an increasing fraction of the space. This is the price paid for the emergent minima property.

5. **Finite support is a double-edged sword.** The same mechanism that creates emergent local minima (overlapping finite support regions) also creates dead zones (regions outside all supports). The paper's strength and weakness are two sides of the same coin.

References

[1] B. Hoover, K. Balasubramanian, D. Krotov, and P. Ram, "Dense Associative Memory with Epanechnikov Energy," in *New Frontiers in Associative Memory* (Workshop at ICLR 2025), 2025. Available at: `https://openreview.net/forum?id=LSR_DenseAM`

[2] D. Krotov and J. J. Hopfield, "Dense associative memory for pattern recognition," *Advances in Neural Information Processing Systems*, vol. 29, 2016.

[3] H. Ramsauer, B. Schäfl, J. Lehner, P. Seidl, M. Widrich, L. Gruber, et al., "Hopfield networks is all you need," in *International Conference on Learning Representations*, 2021.

[4] C. Lucibello and M. Mézard, "Exponential capacity of dense associative memories," *Physical Review Letters*, vol. 132, no. 7, p. 077301, 2024.

[5] V. A. Epanechnikov, "Non-parametric estimation of a multivariate probability density," *Theory of Probability & Its Applications*, vol. 14, no. 1, pp. 153–158, 1969.

[6] M. P. Wand and M. C. Jones, *Kernel Smoothing*, CRC Press, 1994.